

Nathan Donghyun Kim

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EDUCATION

Georgia Institute of Technology (3.8/4.00)

Atlanta, GA

BS in Computer Science

December 2026

- **Dean's List Honors** (all semesters), **Faculty Honors** (Spring 2025)
- **Relevant coursework:** Data Structures & Algorithms | Design & Analysis of Algorithms | Computer Systems & Networks | Database Systems | Computer Organization & Programming | Objects & Design | Numerical Analysis

PROJECTS & EXPERIENCES

Hospital Database Management System — Lead Engineer

Nov 2025 - Dec 2025

- Architected a full stack hospital operations platform through **React.js**, **Node.js**, & **MySQL**, delivering end-to-end functionality and supporting **15+ queries and workflows** through a unified interface.
- Leveraged **MySQL** stored procedures and parameterized queries to enforce clean separation between the JavaScript layer and SQL execution, **eliminating injection vectors** and centralizing database logic for stronger security and maintainability.
- Implemented client and server side validation layers to catch formatting errors early and enforce business rules, reducing unnecessary server calls and improving system reliability.
- Designed modular **React** components styled with **TailwindCSS**, enabling consistent styling and making new features and forms easy to implement.

CSync: A College Application Management Platform | Co-founder & Full Stack Developer

May 2025 - Aug 2025

- Designed and implemented a full stack web application with **React.js**, **Node.js**, **Express.js**, **MongoDB**, **RESTful APIs**, and **Auth0** to help users track and manage college applications, ensuring maintainability and scalability.
- Improved query performance by 50% through rate limiting, middleware refactoring, caching strategies, and asynchronous data fetching.
- Implemented secure user data and authentication flows with **JWT** and **Auth0 Authentication**, developing **RESTful APIs** to streamline front-to-back end communication.
- Developed responsive, reusable **React** + **TailwindCSS** components, standardizing design patterns and ensuring responsiveness and consistency across all platforms.
- Collaborated in a 2-person **Agile** team, using **Git** for version control, deploying to **Render** & **Netlify** with **99% uptime**.

Grid Resilience, Outage, Weather, Emergency Response Lab | Researcher & Developer

Jan 2025 - Present

- Led development for a data processing pipeline in Python to categorize and aggregate raw daily outage data from dozens of heterogeneous providers through a dictionary-driven architecture in **Python**.
- Reduced data parsing errors by 90% and guaranteed consistent schema designs by designing a preprocessing pipeline that standardizes naming conventions and formatting, leveraging regex and fuzzy matching.
- Integrated US census population data with weather service models to visualize ZIP code-level high-risk outage zones through modular **React** components.

Propulsive Landers at Georgia Tech | Guidance, Navigation, & Controls Vice-Lead

Aug 2024 - May 2025

- Led a 5-member team to improve control loop accuracy and runtime by 20% by developing a custom state-space model and LQR in **Python** and **C++**.
- Reduced path-planning computation time and memory required by 50% by **designing a custom hybrid spline algorithm** that integrated key properties of Hermite & Bezier splines for VTOL rockets.
- Led research for and authored a **peer-reviewed AIAA paper** on *Spline-Based Flight Path Planning and Following for Aerial Navigation*, presenting it at the **AIAA Region 2 Conference**.
- Coordinated weekly design reviews and cross-team collaboration to ensure timely project milestones.

WanderSync: Travel Management System | Full-stack Developer & Database Manager

Sep 2024 - Dec 2024

- Managed **Agile** sprint planning for a 6-member team for a collaborative travel management **Android CRUD app** using **Android Studio**, **XML**, **Java**, and **Google Firebase**.
- Planned and integrated a **Google Firebase** database for real-time **user authentication** and database functionality.
- Improved database design and structuring to **increase query efficiency by 25% and improve readability and manageability**.
- Created **XML layouts** and **Java controllers** to implement interactive UI elements that allowed for storing accounts, creating new trip itineraries, and inviting collaborators to certain trips.

SKILLS

Programming Languages

Java | JavaScript | Python | SQL | C | C++ | TypeScript | MATLAB

Technologies & Frameworks

Node.js | React.js | Express.js | TailwindCSS | MySQL | MongoDB | RESTful API | Pandas | JWT & Auth0 Authentication | Cloud Deployment (Render & Netlify) | Git | Docker | Google Firebase | HTML | Figma

PUBLICATIONS

- **Kim, N.**, Sanchir, E., Law, K. *Spline-Based Flight Path Planning and Following for Aerial Navigation*. AIAA, 2025. <https://arc.aiaa.org/doi/10.2514/6.2025-99477>.